

# ECON 0100 | Homework A1 | PPF

## I Solutions

*Answers and work are shown in the red blocks.*

### The Hogsmeade Candy Shop Saga

Honeydukes is a candy shop in Hogsmeade. The shop makes two types of popular candy, Chocolate Frogs ( $F$ ) and Ice Mice ( $M$ ). Devoting all their resources to one good, Honeydukes can make 100 pounds of  $F$  or 75 pounds of  $M$  each week.

#### Q1 | Honeydukes' PPF

- a) Set up Honeydukes' weekly PPF on an  $x, y$  graph with Ice Mice ( $M$ ) on the vertical and Chocolate Frogs ( $F$ ) on the horizontal.
- b) What is Honeydukes' opportunity cost of 1F?

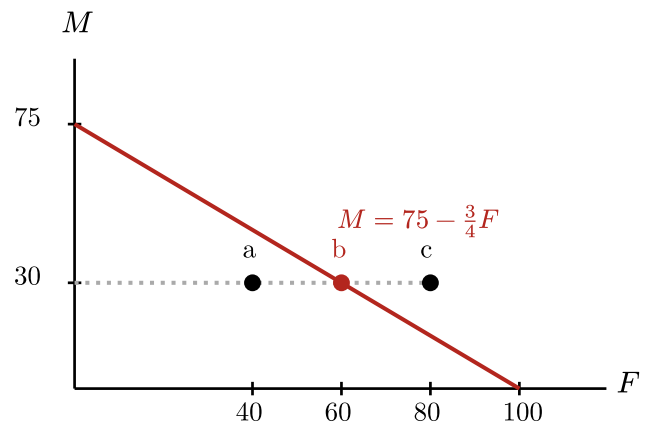
**b)  $\frac{3}{4}$  pound of M.**

**a)** A full week on Ice Mice is the vertical intercept, 75 M; a full week on Chocolate Frogs is the horizontal intercept, 100 F. The PPF is the line between them,  $M = 75 - \frac{3}{4}F$ . The three Q2 orders are marked.

**b)** A week is 100F or 75M, so  $100F = 75M$  and

$$1F = \frac{75}{100}M = \frac{3}{4}M$$

Flip it for the other good:  $1M = \frac{4}{3}F$ . The  $\frac{3}{4}$  is also the absolute slope of the PPF: Ice Mice given up per extra pound of Frogs.



## Q2 | Feasibility

The shop receives three owl-orders for next week. Classify each production plan as inefficient, efficient, or unattainable, using your graph or algebra to justify your answers.

a) 40F and 30M

b) 60F and 30M

c) 80F and 30M

**a) Inefficient. b) Efficient. c) Unattainable.**

**Graph:** all three orders sit on the 30M line. The frontier allows  $M = 75 - \frac{3}{4}F$ , which is 45 at  $F = 40$ , 30 at  $F = 60$ , and 15 at  $F = 80$ . So  $(40, 30)$  is below the frontier,  $(60, 30)$  is on it, and  $(80, 30)$  is above it.

**Algebra (time check):** 30M takes  $\frac{30}{75} = \frac{2}{5}$  of the week in every order. The Frogs take  $\frac{40}{100} = \frac{2}{5}$ ,  $\frac{60}{100} = \frac{3}{5}$ , and  $\frac{80}{100} = \frac{4}{5}$  of the week, for totals of  $\frac{4}{5}$ , 1, and  $\frac{6}{5}$  weeks. Under a full week is inefficient, exactly one week efficient, over one week unattainable. Either justification earns full credit.

### Q3 | Dynamics

Ahead of the holiday rush, Honeydukes hires a second shift, doubling the shop's labor. Right after, their Chocolate Frog mold cracks, cutting their Chocolate Frog productivity in half.

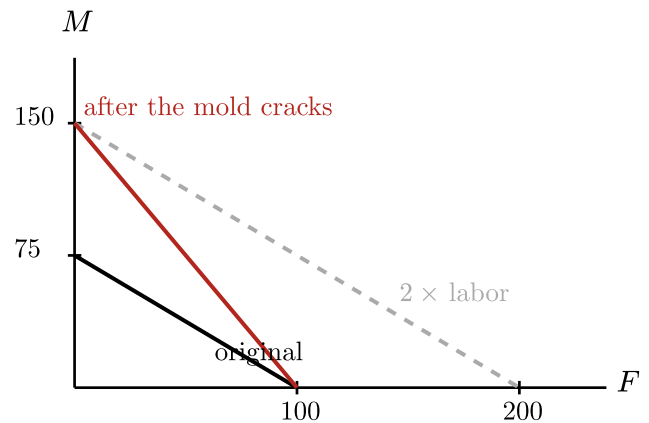
a) Show both changes to Honeydukes' PPF on one graph.

b) After both changes, what is Honeydukes' opportunity cost of 1F? Of 1M?

**b)**  $1F = \frac{3}{2} M$  and  $1M = \frac{2}{3} F$ .

**a)** The second shift doubles both intercepts, a parallel shift out to 200F or 150M. The cracked mold then halves only the Frog intercept, back to 100F, while the Mice intercept stays at 150M: the PPF pivots and its slope changes.

**b)** After both changes a week is  $100F = 150M$ , so  $1F = \frac{150}{100}M = \frac{3}{2}M$  and  $1M = \frac{100}{150}F = \frac{2}{3}F$ . Frogs have become twice as expensive as before ( $\frac{3}{4}M$  to  $\frac{3}{2}M$ ): more labor scales both goods and leaves the trade-off alone, but a change that hits only one good moves it.



## Q4 | Next Best Alternative

Neville visits Honeydukes with one galleon, enough to buy exactly one item: a pound of Chocolate Frogs, a pound of Ice Mice, or a Sugar Quill. He likes Chocolate Frogs best, then Sugar Quills, then Ice Mice.

- a) What is Neville's opportunity cost of buying the Chocolate Frogs?
- b) Honeydukes adds Fizzing Whizzbees to the shelf, also one galleon. Neville likes them more than Sugar Quills but less than Chocolate Frogs. What is his opportunity cost of the Chocolate Frogs now?

**a) The Sugar Quill. b) The Fizzing Whizzbees.**

Opportunity cost is the single next-best alternative, not the sum of everything forgone. Before, the ranking is Frogs > Quill > Mice, so buying the Frogs costs him the Quill; the Mice are not part of it, since he would not have bought them anyway. The Whizzbees slot in between: Frogs > Whizzbees > Quill > Mice. The Frogs are still his best buy, but they now cost him more: the best thing he gives up is the Whizzbees. The common mistake is “the Quill and the Whizzbees”; only one alternative can be next best.